

Feasibility of Implementing a Rain Garden on the UIUC North Engineering Quad

By: Varsha Parthasarathy, Sachi Thakur, Rianna Rane, Lia Zurek

December 12, 2025

Milestone #3

CEE 190

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Introduction

The North Engineering Quad of the University of Illinois at Urbana-Champaign (UIUC) lies between the Nick Holonyak Micro and Nanotechnology Laboratory and Newmark Civil Engineering Laboratory buildings. There is a grass-covered hill at the center of this quad, on which students, professors, and other pedestrians walk to access nearby buildings daily. The surrounding buildings on this quad include the Electrical and Computer Engineering Building, the Civil and Environmental Engineering Building, the Beckman Institute, and Digital Computer Laboratory, making it a heavily utilized area on campus. Due to this heavy foot traffic, the soil of this hill has likely compacted, limiting the infiltration of stormwater, highlighting the importance of ensuring that any stormwater management solutions are designed to function effectively without disrupting daily activities or access for students, faculty, and visitors.

Stormwater can lead to other environmental issues on the quad, including flooding and contamination of nearby water bodies. **Figure 1** shows a section of the quad with stagnant still water after recent precipitation. This pooled water poses a risk for pedestrians slipping and structural damage on the walkways, emphasizing the urgency of this issue.



Figure 1. Pooling water on the North Quad (Photographed by A. Schmidt April 5, 2017).

Historically, flooding has been documented north of the Bardeen Quad and Boneyard Creek, which is a five-minute walk from the North Engineering Quad (Delgado 2024). This is because of the limited capacity of Boneyard Creek to hold heavy rainfall, resulting in flooding and damage to nearby construction projects (University of Illinois Archives 2014). Due to the North Engineering Quad's proximity to Boneyard Creek, where much of the university's stormwater is drained, there have been issues with overflowing and clogged drainage systems on campus (Champaign County History 2024). Lastly, as shown in **Figure 2**, the greenspace on the North Quad is primarily turf, with low variety of flora. This limits plant biodiversity.



Figure 2. South-facing view of the North Engineering Quad at the University of Illinois Urbana-Champaign, featuring Newmark, the CEEB, the ECEB, and the Grainger Library (Google Maps 2025).

The quad currently lacks sufficient runoff management techniques to capture the amount of rainfall it receives. This can result in flooding, higher rates of erosion, and standing water which can result in algal blooms that ultimately create dead zones in the water and eliminate biodiversity (EPA 2025) (Mezler 2023). There are a few options that could be used to solve this problem. One potential solution is replacing the entirety of the North Quad with permeable pavement like porous asphalt in place of the current nonpermeable pavement. This would help with large scale infiltration on the north of campus since the new pavement would cover a significant area of the quad. Possible interferences of this rain garden include site challenges like plant overhydration, pollutant overload.

The next solution for stormwater management would include a system of permeable interlocking pavers which allows the water to seep through the cracks in the pavement and infiltrate into the ground. Although this solution would help the stormwater management issue, it would result in high levels of disruption to the quad for construction, just like replacing the pavement material. This quad houses many departmental engineering buildings that students and faculty need to access every day. Additionally, most permeable pavements can experience structural problems when it comes to highly trafficked areas (Sustainability Directory 2024). A space consistently used by students and professors must be able to handle this level of foot traffic.

Another way to manage stormwater runoff is the implementation of rain gardens. Rain gardens allow stormwater to infiltrate through permeable soil and plants, preventing a large portion of the water from entering the sewer system and reducing runoff. One study from the University of California shows that compared to traditional lawns, one rain garden allows for 30% more water to infiltrate into the ground (Gonzalez 2025). A rain garden would likely be a more sustainable option for managing the stormwater on the quad, while also improving the biodiversity of the local ecosystem. The proposed rain garden will also be a good resource for adding native plants and pollinator-friendly flora to an area and can help

improve the aesthetics of the quad. The construction of a rain garden would not highly interfere with the regular users of the North Quad, as it would be implemented at a selected location between the grassy areas and paved paths. Compacted grass hills in the middle would be ineffective for the proposed rain garden since it would only capture direct precipitation during rainfall and not collect excess runoff after the fall. If the rain garden was on the hill, the domed structure of the hills would cause the runoff to flow onto the nonpermeable sidewalk pavement, which would be counterproductive. Implementing a rain garden on the North Quad will reduce flooding, decrease peak stormwater, runoff volumes, and increase the aesthetic value of the area.

Objectives

The goal of this study was to determine if a rain garden is a feasible investment on UIUC's North Engineering Quad and to establish the environmental and financial implications of this proposed solution. Construction, installation, maintenance, and blueprinting of the rain garden were not executed in this study. The completion of this study enabled proper future measures to be taken on the North Quad for stormwater management, with the understanding that these measures would positively impact flood risk and be financially reasonable. Pollinator-friendly plants were also considered in the research. A final report and presentation with all completed assessments, data collections, and a concluding feasibility statement were delivered by December 12, 2025.

Methodology

This feasibility study of adding a rain garden on the North Engineering Quad began with a literature review of past similar projects to create a general overview and set goals for this project. A site assessment was then conducted, with a focus on the land properties of the possible sites. After the data collection, a parametric cost analysis was completed using the literature review to approximate a final cost estimate of the rain garden. The margins for the trees and sidewalks were also considered when finding the total area for the rain garden. Finally, a concluding feasibility report which summarized the findings and included a statement of feasibility was completed.

Literature Review and Data Collection

Case studies of rain gardens of similar scale were reviewed, specifically those implemented on other college campuses. Precipitation data of the Champaign area along with maps of current stormwater management infrastructure on the North Quad were also considered.

Site Visit and Location Analysis

A site visit to the North Engineering Quad was conducted on November 3, 2025. The distance from each plot to the center of the hill and the slope of the hill were measured with a standard measuring tape. Next, the dimensions and qualitative observations of any unique features for each location were noted.

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Post Site Discovery Research

After the site visit, pedestrian data near each site was collected, by a physical study of counting the number of pedestrians on each sidewalk during a high traffic time between classes. This was completed with the goal of understanding the impact of implementing such a project on pedestrians on the quad. For this data, pedestrians are defined as individual people walking or biking on the sidewalk. The center of the bottom half of the quad was used as a reference point (R.F. 3) to count the number of pedestrians walking on each sidewalk during a 20-minute period (2:45 PM - 3:05 PM) on November 11, 2025. The reference point and analyzed sidewalks are shown in **Figure 3**. Additionally, research on plants native to Central Illinois was conducted to ensure an increase in biodiversity on the quad through the rain garden.



Figure 3. Map of North Quad with reference points and sidewalks used for pedestrian data. (Google Maps 2025)

Site Selection and Preliminary Design

Pedestrian data and stormwater management data were used to select a site. Dimensions of the selected site, along with the addition of margins for the sidewalk and circular margins for preexisting trees were utilized to create a preliminary site design of the proposed project. The area of the proposed rain garden was then calculated from this design.

Cost Estimation

Project cost data was extracted from the case studies previously reviewed. This data in addition to the proposed area of the rain garden was used to create a parametric cost analysis by finding a scaled cost from each case study. These scaled costs were then averaged to find the projected cost of the project.

Cost Benefit Analysis

The pedestrian data and Green Values Calculator were used to understand stormwater management benefits and interference with regular activity. Other benefits, such as aesthetic and environmental benefits of additional biodiversity were also considered. All factors and cost estimation were reviewed to determine the overall feasibility of the proposed project.

Report and Presentation of Findings

A report of all the collected data, research, calculations, and findings was compiled. In addition, a presentation was made to easily communicate findings to an audience. The report and presentation were submitted and presented for review.

Results and Discussion

Over the past 30 years, the North Engineering Quad at the University of Illinois Urbana-Champaign has received an average monthly rainfall of approximately 3.3 inches. **Table 1** shows the comparison of the 2025 and 30-year average rainfall. In 2025, the average monthly rainfall was 1.94 inches based on data collected from January through October of this year so far. The months with the highest rainfall averages are April through August. These months are consistent with both the 30-year average and the 2025 average rainfalls. The 30-year average rainfall numbers were computed by taking the average of the precipitation in inches of each month from 1970 to 2000 (Precipitation 2025). A rain garden requires approximately 1 to 1.5 inches of rainfall to achieve significant infiltration effects. **Table 1** indicates that seven out of twelve months in 2025 met the criteria for a productive rain garden. Additionally,, the current conditions of Champaign, Illinois have been classified as a drought, and with over half of the 2025 values larger than this 1-inch margin, there is a valid risk of high flow of runoff even in such drought conditions (Drought.gov 2025). This substantiates that the North Quad meets the regional rainfall conditions necessary for proper rain garden performance (Illinois Department of Natural Resources 2025).

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Table 1. 30 Year Average of Rainfall Data on North Engineering Quad. (Precipitation 2025)

Month	30-yr avg	2025	Difference
January	2.23"	0.83"	-63%
February	2.15"	0.59"	-73%
March	2.73"	2.51"	-8%
April	4.02"	3.63"	-10%
May	4.49"	2.86"	-36%
June	4.64"	4.94"	+6%
July	4.23"	4.44"	+5%
August	3.28"	0.53"	-84%
September	3.20"	1.33"	-58%
October	3.33"	1.43"	-57%
November	3.12"	N/A	N/A
December	2.29"	N/A	N/A

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Table 2 displays photos of the three locations selected with description of the observations noted.

Table 2. Descriptions and photographs of each location, taken November 3, 2025.

	Description	Photo
Location 1	Location 1 on the North Engineering Quad is between the Nick Holonyak Micro and Nanotechnology Laboratory and the upper section of the compacted hill. The calculated area of this plot is 424.5 ft ² . A notable feature of Location 1 is a fire hydrant close to the sidewalk. This poses a risk of interference with the installation and may limit the area of the potential rain garden. If this location is selected, a thorough examination of the entire hydrant system must be conducted.	
Location 2	Location 2 also lies alongside the Nick Holonyak Micro and Nanotechnology Laboratory. It has the largest plot of the three locations with an area of 473.7 ft ² . Near the plot, there is a roof drain overflow pipe attached to the building. However, this will not greatly impact the runoff rates towards the rain garden outside of extreme cases.	
Location 3	Finally, Location 3 is on the East side of the North Quad right outside of the entrance to the Newmark Civil Engineering Laboratory. It is the smallest plot at 421.7 ft ² . There are several bushes lining the building that would sit behind the rain garden.	

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Table 3 displays the data collected during the measurement process, including the measured dimensions, calculated areas, and calculated slope of the hill in the center of the quad.

Table 3. North Engineering Quad data collected on November 3, 2025.

	Distance from quad to sidewalk (m)	Length (N/S) of plot (m)	Width (E/W) of plot (m)	Calculated area of plot (m ²)	Calculated area of plot (ft ²)	Calculated slope of the hill (m)
Location 1	10.2	8.00	4.93	39.4	424.5	0.093
Location 2	10.6	8.48	5.19	44.0	473.7	0.093
Location 3	11.7	9.09	4.31	39.8	421.7	0.093

After organizing the data collected on each location, the calculated average area of all three plots is around 440 ft². Since each location had a tree on each side of the plot, to ensure the preservation of these trees and to decrease the risk of interference of outside factors on the rain garden, a design of the proposed rain garden with a margin for the trees was created, as shown in **Figure 4**. The border of the margin for each tree is presented in the blue dashed line, with a 7 ft margin radius from the tree. With this margin, the final area of the proposed rain garden to be used in the cost estimation totals to 290 ft².



Figure 4. Design of rain garden with a margin for trees.

Figure 5 shows a map of the storm sewers on the North Engineering Quad. The three observed locations are circled in red, matching the labels previously presented in **Figure 2**. On the map, the small colored circles mark the sewer drains on the surface of the group, and the lines represent the paths of the sewer pipes throughout the quad. Location 1 already has some forms of stormwater management, with three sewer drains marked in the figure. Although Location 2 doesn't appear to have any surface level sewer drains, there is a nearby sewer line which runs on the West side of the plot, close to the Nick Holonyak Micro and Nanotechnology Laboratory. However, the area on and around Location 3 has no marked sewer drains or pipes, signaling this location to have a more urgent need for a rain garden to supplement as a sufficient stormwater management system.

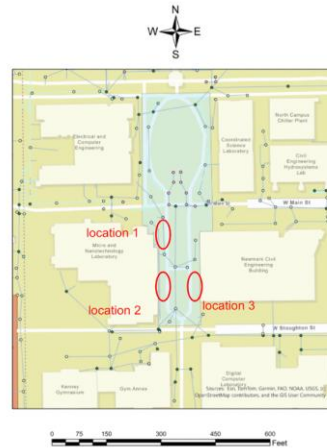


Figure 5. Map of the storm sewers on UIUC's North Engineering Quad, provided by A. Schmidt, University of Illinois (Personal communication, Nov 12, 2025).

Table 4 highlights some plants to be used in a potential rain garden that are both native to the Central Illinois Region and best suited for rain gardens. This information was identified to clarify additional feasibility benefits of the study, as it will be useful to note if the rain garden is confirmed to be feasible.

Table 4. Plants native to Central Illinois and UIUC areas. (College of Agricultural, Consumer & Environmental Sciences 2020) (Illinois Department of Natural Resources 2025)

Native Plant	Benefit
Swamp milkweed (<i>Asclepias incarnata</i>)	Good choice for basin zone, tolerates flooding and handles dry conditions well
Lobelia siphilitica (Great Blue Lobelia)	Bright purple color, tolerates wet-mesic soils and partial sun
Blue vervain (<i>Verbena hastata</i>)	Tolerates moist to wet soils and works well w pollinators.
Prairie Dropseed (<i>Sporobolus heterolepis</i>)	Suited for edge/ bank of rain garden – visually appealing.

Contacted facilities and services were not able to provide pedestrian data on the North Engineering Quad. To include interference with daily student traffic on the Quad and the rain garden, a pedestrian count was conducted as a part of this study in place of the actual data. This count also observed that many students use the hill for events and programs, meaning the hill serves as a crucial part of the quad for daily activity. The pedestrian counts shown in **Table 5** demonstrate a notable difference between the pedestrian traffic on the sidewalks. Location 3 lies on the East sidewalk and Locations 1 and 2 lie on the West sidewalk. From the data below, the East sidewalk receives far less pedestrian traffic than the West sidewalk. To reduce the interference of regular activity for students and professors, it would be more effective to place the proposed rain garden in a location with the least amount of pedestrian traffic.

Table 5. Pedestrian Data from North Quad sidewalks.

	Pedestrian Count
East Sidewalk	32
West Sidewalk	45

The three different locations selected at the North Engineering Quad have an average area of about 440 ft². To prevent inconvenience and interference with other plants, margins around the sidewalk and margins around the trees on the site were removed from the total area of the rain garden. This brings the total area of the proposed rain garden to approximately 290 ft². This cost estimate considers the cost of labor but does not include the cost of maintenance. By utilizing the case studies from Glendale Public School, UIUC, and the University of Wisconsin in the parametric cost estimation, the projected cost of the proposed rain garden is calculated to be approximately \$2300, as shown in **Table 6**.

Table 6. Parametric Cost Estimation of Rain Garden.

Project	Cost	Area	Cost/Area	Scaled Cost
Glendale Public School Rain Garden (Sustainable Technologies 2020)	\$128,563	15,338.57 ft ²	\$8.38/ft ²	\$ 2,429.95
UIUC (Tarpey et. al 2017)	\$10,618.88	1,629.65 ft ²	\$6.52/ft ²	\$1,890.60
University of Wisconsin (UW Applied Ecological Services 2007)	\$1,862.5	200 ft ²	\$9.30/ft ²	\$2,696.72
Projected Cost				\$2,339.09

The Green Values Calculator was used to understand the effectiveness of implementing a rain garden onto the North Quad. This is done by analyzing the overall site, identifying which sections of the site contain pervious and impervious pavements. Then, assigning square footage to each of the sections and identifying which sections will be converted into a rain garden. For the proposed project, half of the hill, the nearby sidewalks, and the outside margins of the quad were used for the Green Values Calculator (Center for Neighborhood Technology 2025). According to the calculator, a 290 ft² rain garden on the outside margins of the quad will capture 100% of the water volume that would runoff from the hill and nearby sidewalks. In addition to this, it was shown that a rain garden would increase the water volume capture capacity by roughly 245 ft³. The exact results from the Green Values Calculator are shown in **Figure 6**.

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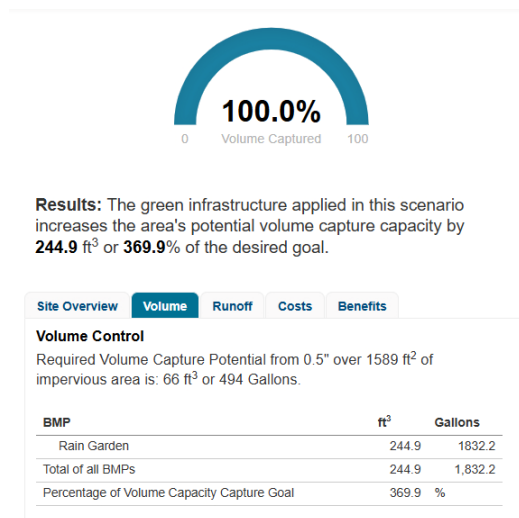


Figure 6. Green Values Calculator results (Center for Neighborhood Technology 2025).

Conclusions

The grass-covered hill in the center of UIUC's North Engineering Quad has likely been compacted by heavy foot traffic, risking low stormwater infiltration and high runoff rates. This can also cause compacted soil, which results in standing water on the quad. This project studied the feasibility of placing a rain garden in one of the side plots, between the sidewalks and neighboring buildings, on the quad to collect runoff. Out of the three locations observed, Location 3, the plot of land between the Newmark Civil Engineering Laboratory and the East side of the North Engineering Quad, lacked sufficient drainage systems compared to the other two locations. This plot measured approximately 9.09 m in length and 4.31 m in width, with a total area of 39.2 m² or 422 ft². Additionally, there are no visible interfering structures nearby, meaning that Location 3 is a more feasible location for the rain garden. Including a 7-foot circular margin around the trees, the area of the proposed rain garden is 290 ft². Through a parametric cost estimation using similar rain garden projects in Canada, Wisconsin, and UIUC itself, a 290 ft² rain garden would cost the University approximately \$2,300. Additionally, the Green Values Calculator showed that a 290 ft² rain garden placed on Location 3 would allow over 1,800 additional gallons of water to infiltrate naturally into the ground. This shows that the addition of a rain garden to Location 3 would be highly effective at stormwater management. This low cost combined with the high effectiveness of stormwater management and aesthetic values, concludes the findings that a rain garden on the North Engineering Quad is feasible.

Acknowledgements

This feasibility study was made possible through the guidance and support of Professors Schmidt, Henschen, and Roesler. We appreciate our CEE 495 mentors, Vishnu and Jovany, for their valuable suggestions on our report and slide deck throughout the semester. Their expertise in managing and executing group projects provided an insightful and essential contribution to the project. We also thank Beth Leitz, Justin Pinnel, and Morgan White from the University of Illinois Facilities & Services for providing storm sewer data for the North Quad which was needed for completing this analysis.

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Group reflections

Sachi Thakur:

This CEE 190 project with my group was very intellectual since I learned a lot about how engineering is applied in the real world and how there are many aspects of a project that need to be considered for CEE initiatives since they directly impact the community. This project included many pre-work tasks that needed to be done before conducting the feasibility study. Decisions were made on how to collect data throughout the project. This includes how to conduct the outdoor measurements of each rain-garden location, how and where to receive pedestrian data from, and how to construct our parametric cost analysis so it can be as accurate as possible. Presentation, data collection, and formal reporting are some skills that our team gained and built on. We learned that when considering earth projects, it is crucial to consider how they can impact trees since they are a nonrenewable resource, and damage to underground roots can restrict tree growth.

Some advice I would give the CEE 190 students is to check over work a lot since one silly mistake can lose a lot of points on a practice project like this. I would also recommend asking many questions since there is so much knowledge to be learned as a new student in the CEE field. Lastly, the CEE 495 mentors gave us meaningful advice that helped us improve our previous milestone reports to strengthen them for the future.

Varsha Parthasarathy:

Throughout this semester, I have learned a lot about technical writing, teamwork, and time management. I especially enjoyed learning about all of the aspects that go into creating a successful project, such as doing physical studies, research, and creating figures and tables. One part of this project that my group struggled with was finding data, especially of our chosen location. There was not much pedestrian data available online, so we decided to do a physical study and count the number of pedestrians on each sidewalk during a high traffic time period. From this, my group learned that although there may be roadblocks along the way, looking for creative solutions, and getting things done ourselves is always a good and effective option. Throughout the semester, my team also improved our technical writing skills. We learned how to write to communicate ideas effectively and efficiently, without extra or unnecessary details. This skill will be very useful in our futures, as environmental engineers must do a lot of technical writing in their jobs. For future CEE190 students, I would highly recommend going to professor office hours and asking them as many questions as you can. This is especially helpful because the professors have final say over your grade, so getting their advice is the most useful for doing well on assignments.

The CEE495 peer mentors and the instructor mentors had a big impact on our project. The peer mentors helped us a lot by providing us their past experiences with the class and would give us advice on the small questions that we had. The instructor mentors were very useful for asking specific questions about aspects of our report. When it came to writing the milestones, and questions about how to improve our technical writing or formatting, going to the professor's office hours were very useful.

Lia Zurek:

The CEE190 team project assisted my learning of the benefits and difficulties of collaborative work, long-term dedication to one assignment, and the importance of time-management. It was an interesting effort to explore how to divide the workload of this project in a way that played to each member's strengths and worked with everyone's schedule. At times, it was difficult to communicate or schedule meetings when one or several team members had more stressful weeks or work that required attention in other classes. However, as the semester progressed, we got a better understanding of how to manage our time and ensure that even amongst other stress, this project remained a priority. Additionally, we began to be more comfortable asking for help from our professors. This change corresponded with better understanding of what was being asked of us in each segment of the project and how to proceed. As beginners in the engineering world, this project really challenged all of us to be creative and learn firsthand what sort of information we'd need for our topic. For the next class of CEE190 students, I'd highly recommend being in a group that you get along well with but also can rely on to complete their portion of each assignment. Moreover, I'd tell them how helpful office hours can be to further their understanding of what needs to be done for the project.

The CEE495 mentors were a great way to get feedback on our project in a way that felt less intimidating than asking a professor or a TA. Knowing that the mentors had been in the class just a few semesters prior made it feel like they knew exactly what we were experiencing with the project and could give very relevant advice. Instructor mentoring was vital to our project. Going to Professor Schmidt's office hours whenever possible to ask questions about grading our about how we can improve our draft gave us a very clear idea of how to proceed and refine our project.

Rianna Rane:

Throughout our CEE 190 project, my group went through a rollercoaster of decisions, failures, and stressful moments, which helped reveal our strengths and weaknesses as individuals and as a group together. From each of our own discussion assignments to our group Milestones, our ideas and technical writing styles intertwined to create a final project which represented a collective work cumulated over the semester. Personally, I began the semester feeling a little lost while writing the discussion assignments, unsure of which content I should include and how I could present it in a technical report. What initially felt impossible to write "well" slowly became a project which I enjoyed watching progress into a compilation of all our ideas. One factor I did not foresee struggling with was the mixing of all our writing styles. Since technical writing focuses on the intention and analytical details, each of us wrote with our own twist of character into it, sometimes making the writing feel choppy. I am very proud of our team for working on this weakness and being able to truly blend each other's' styles into our final project. My advice to future CEE 190 students is to remember to never fear the "work" that you may have to do but learn to foster it as it progresses.

The CEE 495 mentors gave us numerous valuable insights on not only our project, but also our future studying CEE at UIUC. We began every mentoring hour by having them review any progress we had made on our report or presentation. Their advice became increasingly helpful as we, ourselves, became more familiar with the project and what we envisioned our end goal to be like. In addition to feedback on our project, the mentors would also show or talk about their CEE 190 project from the past. One of the most helpful things they did was give us advice on our future class plans for the next 4 years. From telling us which class workloads would be overwhelming in the same semester to describing the skills we learned in certain CEE classes, I feel more prepared and knowledgeable about my future path.